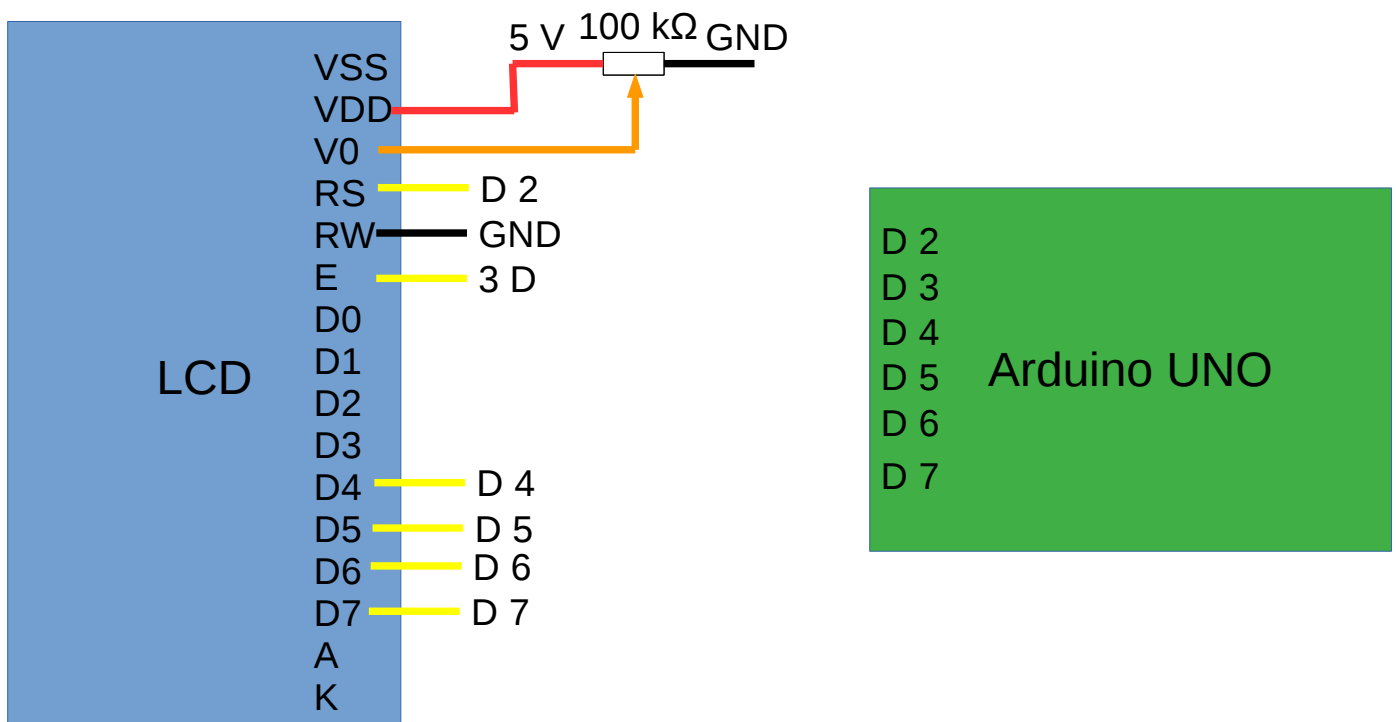
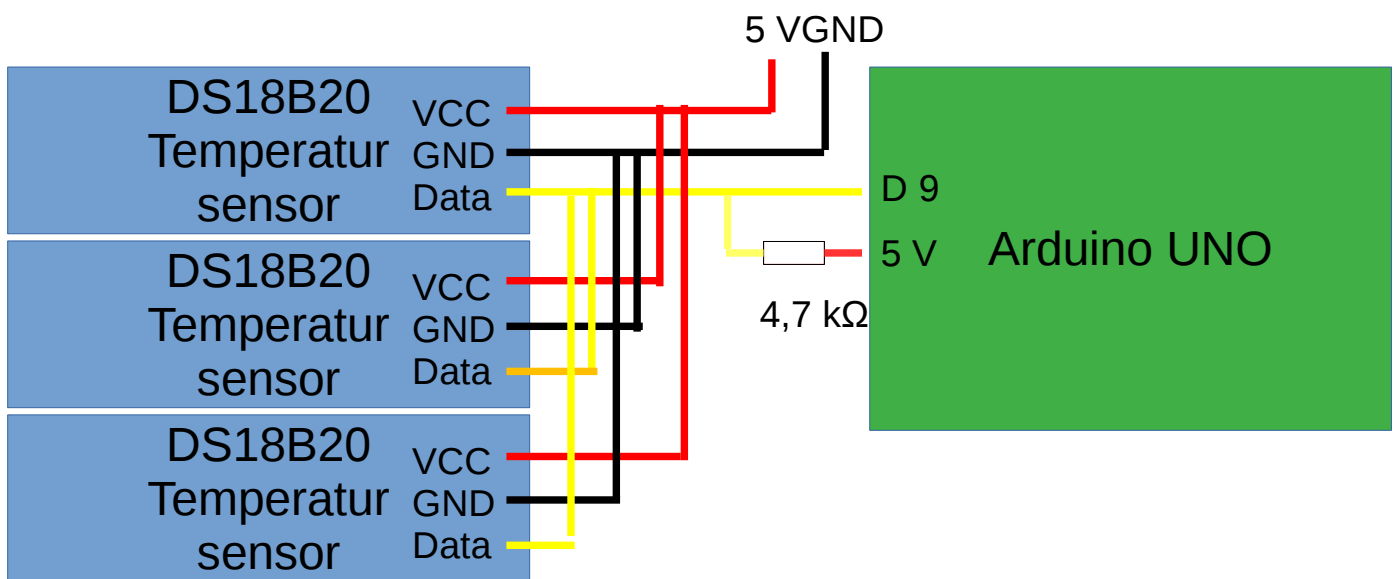
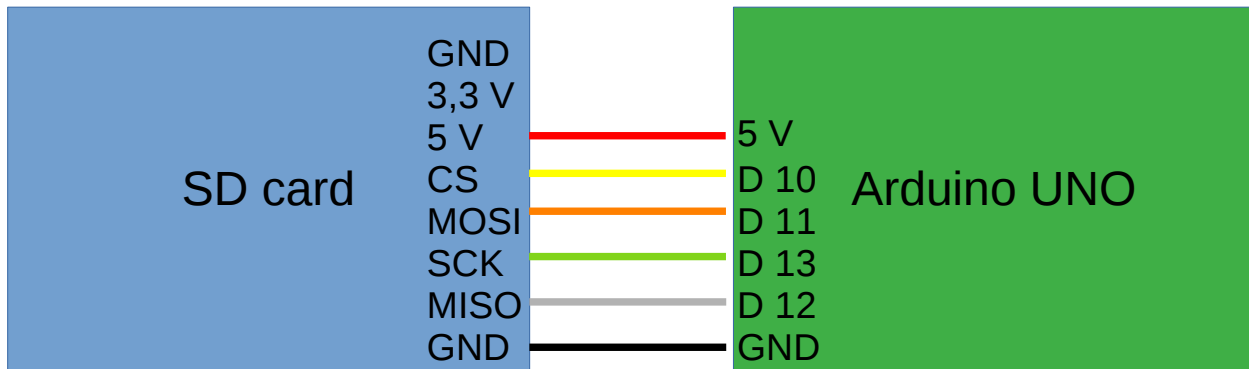


Esquema conexión datalogger

sketch temp_logger_v250918.ino



```
/*
 * Temperature logger with
 * multiple temperature sensors DS18B20
 * SD card module from lctech (www.lctech-inc.com) -works only with original Arduino not with clone
 * Temp ssensors on pin 9
 * Sd card data (CS ) on pin 10
 */
```

```
#include <OneWire.h>          // DS18B20 temp sensor library
#include <DallasTemperature.h> // DS18B20 temp sensor library
```

```
#include <SPI.h>              // SD card module library
#include <SD.h>               // SD card module library
```

```
#include <LiquidCrystal.h>    // LCD module library code
```

```
File dataLog;
boolean sd_ok = 0;
int loopcount = 0;
int sensor = 0;          // identifies each sensor
int dl = 2000;           // delay time 2000 = 2 s
int time_count = 0;      // time passed since sketch start
```

```
// DS18B20 temp sensor data wire is conntec to the Arduino digital pin 9
#define ONE_WIRE_BUS 9
```

```
// Setup a oneWire instance to communicate with any OneWire devices
OneWire oneWire(ONE_WIRE_BUS);
```

```
// Pass our oneWire reference to Dallas Temperature sensor
DallasTemperature sensors(&oneWire);
```

```
// LCD - initialize the library with the numbers of the interface pins
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
```

```
int numberOfDevices; // Number of temperature devices found
```

```
DeviceAddress tempDeviceAddress; // We'll use this variable to store a found device address
```

```

void setup() {

    // open serial communications
    Serial.begin(9600);

    /*******temp sensor setup start*****

    // Start up the library
    sensors.begin();

    // Grab a count of devices on the wire
    numberOfDevices = sensors.getDeviceCount();

    // locate devices on the bus
    Serial.print("Locating devices...");
    Serial.print("Found ");
    Serial.print(numberOfDevices, DEC);
    Serial.println(" devices.");

    // Loop through each device, print out address
    for(int i=0;i<numberOfDevices; i++) {
        // Search the wire for address
        if(sensors.getAddress(tempDeviceAddress, i)) {
            Serial.print("Found device ");
            Serial.print(i, DEC);
            Serial.print(" with address: ");
            printAddress(tempDeviceAddress);
            Serial.println();
        } else {
            Serial.print("Found ghost device at ");
            Serial.print(i, DEC);
            Serial.print(" but could not detect address. Check power and cabling");
        } // else
    } // for
    /*******temp sensor setup end*****

    /*******SD card setup start*****

    // initialize the SD card
    Serial.print("Initializing SD card...");

    if (!SD.begin(10))
    {
        Serial.println("initialization failed!");
        while (1);

    }
    else
    { // initialization OK
        sd_ok = 1;
        Serial.println("initialization done.");
        if( SD.exists("Log.txt") == 0 ) // test if file with name 'Log.txt' already exists
        { // create a text file named 'Log.txt'
            Serial.print("\r\nCreate 'Log.txt' file ... ");
            dataLog = SD.open("Log.txt", FILE_WRITE); // create (&open) file Log.txt
            if(dataLog) { // if the file opened okay, write to it:
                Serial.println("OK");
                // write text to 'Log.txt' file
                dataLog.println("TEMPERATURE");
                dataLog.println();
                dataLog.close(); // close the file

            }
            else
                Serial.println("error creating file.");
        }
    }

    Serial.println("TEMPERATURE");
    /*******SD card setup end*****

    /******* LCD setup start*****
    lcd.begin(16, 2); // set up the LCD's number of columns and rows:
    /******* LCD setup end

} //(finish setup)

```

```

void loop()
{
    // Call sensors.requestTemperatures() to issue a global temperature and Requests to all devices on the bus
    sensors.requestTemperatures();

    // print data on PC serial monitor*****

    // Loop through each device, print data on PC serial monitor
    for(int i=0;i<numberOfDevices; i++) {
        Serial.print(sensors.getTempCByIndex(i));
        Serial.print(";");
    }

    Serial.print("time = ");
    Serial.print(time_count);
    Serial.print(" s");
    Serial.println();

    // write data to SD card*****
    if(sd_ok)
    { // if the SD card was successfully initialized
        // open Log.txt file with write permission
        dataLog = SD.open("Log.txt", FILE_WRITE);

        for(int i=0;i<numberOfDevices; i++) {
            dataLog.print(sensors.getTempCByIndex(i));
            dataLog.print(";");

        }
        dataLog.print("time = ");
        dataLog.print(time_count);
        dataLog.print(" s");
        dataLog.println();

        dataLog.close(); // close the file
    }

    // write data to LCD *****
    for(int i=0;i<numberOfDevices; i++) {
        lcd.clear(); //Clears the LCD screen and positions the cursor in the upper-left corner.
        lcd.print("sen "); // Print a message to the LCD.
        lcd.print(sensor);
        lcd.print(" = ");
        lcd.print(sensors.getTempCByIndex(sensor)); // Print temp sensor 0 to the LCD
        lcd.print(" C");
        sensor = sensor + 1;
        delay(dl); // wait 2 sec showing sensor value on LCD
    }

    time_count = loopcount * numberOfDevices * (dl / 1000) ;
    loopcount = loopcount + 1;

    sensor = 0;

}

// function to print a device address
void printAddress(DeviceAddress deviceAddress) {
    for (uint8_t i = 0; i < 8; i++) {
        if (deviceAddress[i] < 16) Serial.print("0");
        Serial.print(deviceAddress[i], HEX);
    }
}

```